

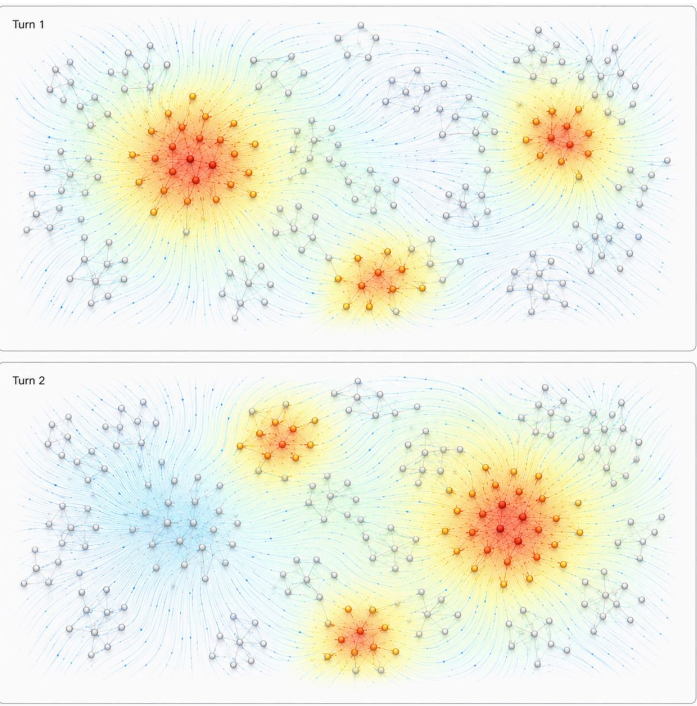
Figure 1 **Reading the ‘Terrain’ Before the ‘Map’ Is Printed** Substrate-level observability and pre-behavioral interventions in multi-agent AI systems

1.a From static map to semantic zoom



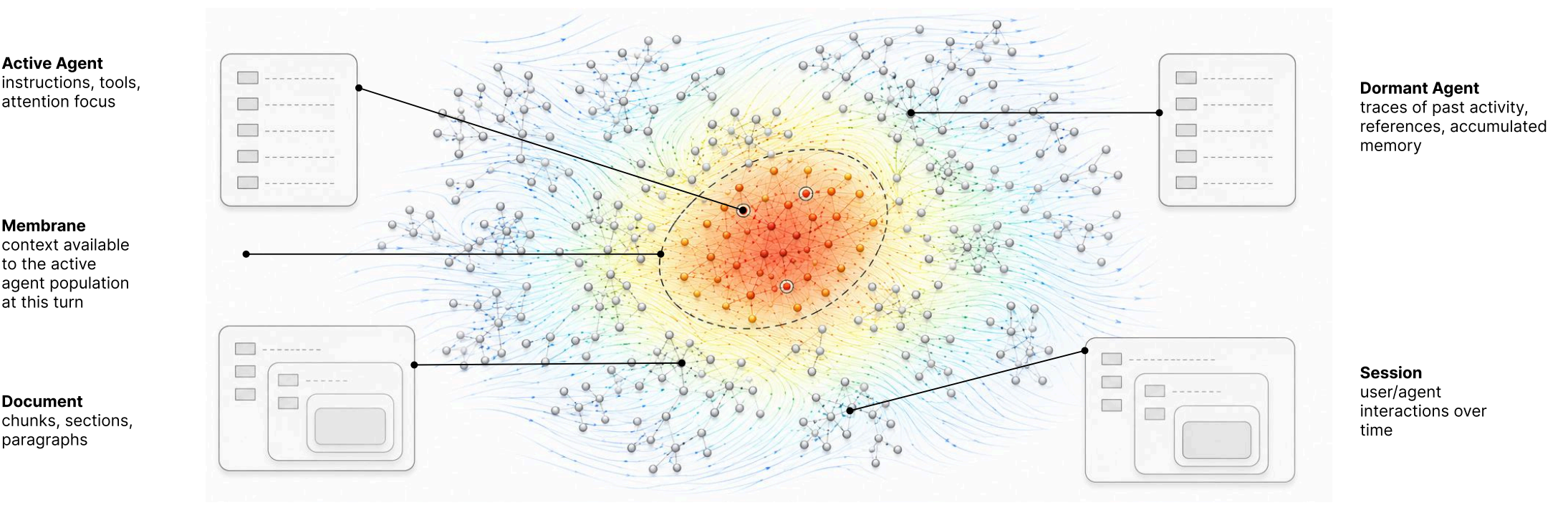
Printed “maps” fix 2D semantic context resolution at publication date; digital maps added 3D zoom with progressive disclosure. The substrate generalizes this: retrieval depth rather than document boundaries – determines what surfaces and at what granularity. A reviewer agent requesting a summary stays at the top; a research agent tracing an argument chain descends into the details.

1.c Phase dynamics: substrate evolution over time

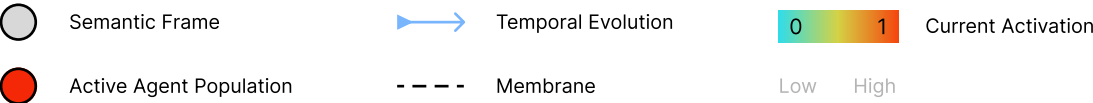


From discrete retrieval to continuous territory:
In flat vector stores, retrieval is discrete: each agent query draws from an undifferentiated pool. The substrate is continuous – shaped by co-retrieval over time. Each access leaves a trace; accumulated traces form territories of heightened salience that persist across turns and bias future retrieval. This is what makes substrate-level observation possible.

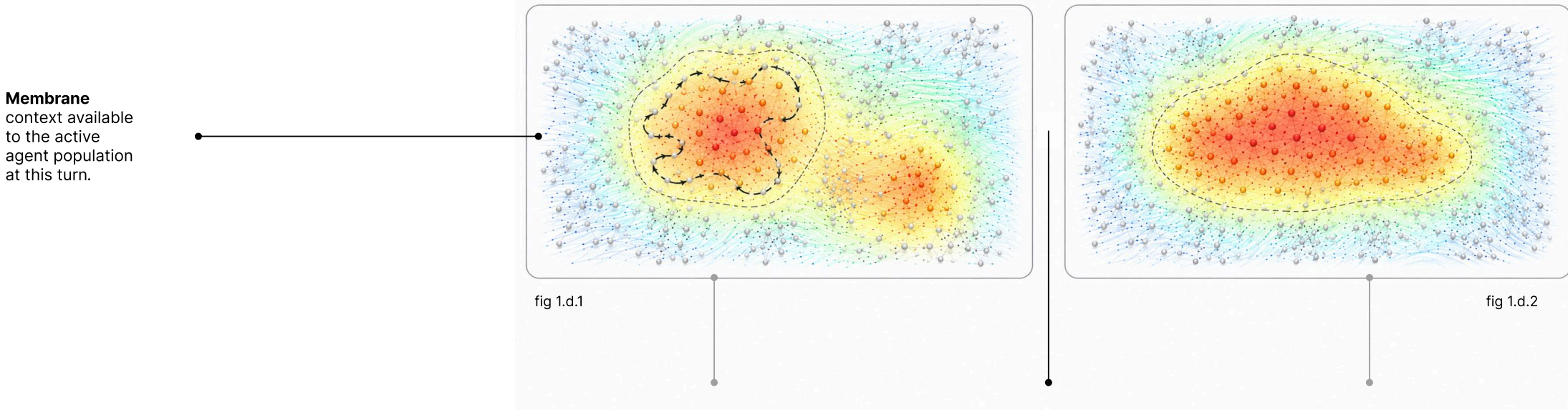
1.b Semantic frame graph and substrate co-retrieval dynamics



Every entity on the substrate – document, agent, session, argument – is a typed frame: a structured set of metadata slots that describe what content the frame holds. These metadata slots may be filled by other frames. This gives us composability: Frames for “Agent state” and “document content” are structurally equivalent hierarchies composed across scales. This self-similarity is what lets substrate signals be measured the same way at every orchestration depth.



1.d Anatomy of a substrate-level intervention



Membrane intervention surface:
The membrane defines the context available to the active agent population at each turn. Interventions adjust position and permeability – modulating flow to redirect co-activation, density to amplify or attenuate regions.

Pre intervention (Recurring attractor)
The active agent population is locked in a narrow region of the substrate: the same frames recur turn after turn, retrieved together, reinforcing one another’s salience. Exploration has stopped — new content is not being pulled into context. At the substrate level this appears as a tight, persistent activation cluster (circular reasoning, deliberative deadlock, or topic-trap construction).

Intervention (membrane expansion)
Substrate-level recurrence patterns — frames retrieved repeatedly across turns without new co-activation — trigger the membrane operator. The membrane shifts position outward and increases permeability between the current activation region and adjacent substrate territory. Flow opens toward previously inactive frames; the cluster begins to spread across the new boundary.

Post Intervention
New frames enter the active context, drawn from the newly accessible substrate region. Co-activation redistributes across the expanded territory; the previously dominant attractor weakens as its frames lose their reinforcement loop. Exploration resumes through previously unvisited substrate regions. The system recovers.